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Precision Agriculture: AI-Based Fruit Yield Estimation System

Bachelors of Science in Computer Science

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Abstract

Agriculture is a cornerstone of Pakistan's economy, contributing significantly to employment, rural livelihoods, and national food security. Orchard crops, particularly apples, play an important role in the agricultural output of northern regions such as Balochistan, Khyber Pakhtunkhwa, and Gilgit-Baltistan. Despite their economic importance, yield estimation practices in Pakistani apple orchards remain largely traditional, relying on manual field surveys, visual inspection, and farmer intuition. These methods are labor-intensive, time-consuming, and often inaccurate, especially in large orchards where tree-to-tree variability is high. The lack of reliable pre-harvest yield estimates limits effective farm planning, market forecasting, storage management, and supply chain coordination, ultimately affecting farmer income and agricultural efficiency.

Recent advancements in artificial intelligence and computer vision provide new opportunities to modernize yield estimation through image-based analysis. However, many existing technological solutions depend on expensive infrastructure such as drones, satellite imagery, environmental sensors, or long-term historical yield records. Such requirements make these systems impractical for widespread adoption in Pakistan, where small- to medium-scale farmers often have limited access to advanced resources. There is therefore a strong need for a low-cost, accessible, and scalable yield estimation system that can operate using minimal equipment while remaining effective under real-world orchard conditions.

This project presents an AI-based fruit yield estimation system designed specifically for apple orchards using only ground-level RGB images captured with commonly available devices such as smartphones. The proposed system integrates multiple computer vision and deep learning components into a unified pipeline that estimates orchard-level yield without relying on aerial imagery or local yield ground truth. The framework consists of four core modules: fruit detection and counting, fruit ripeness classification, canopy feature extraction, and yield estimation.

The fruit detection module employs deep learning-based object detection techniques to localize and count apples in ground-level orchard images, addressing challenges such as occlusion, overlapping fruits, and variable lighting conditions commonly encountered in Pakistani orchards. Detected fruits are further analyzed by a ripeness classification module that categorizes them into discrete maturity stages based on visual color and texture cues. This information provides valuable insights into harvest readiness and supports more informed yield interpretation.

In parallel, the system extracts canopy-related features from ground-level images through image segmentation and geometric analysis. Canopy indicators such as projected area and spread serve as proxies for tree vigor and structural capacity, complementing fruit count information. These visual cues are then combined using a literature-informed yield estimation model that approximates fruit yield by integrating fruit quantity, ripeness distribution, and canopy characteristics, without requiring locally collected yield labels.

To ensure practical usability, the system is integrated into a farmer-friendly mobile dashboard that allows users to capture images, view intermediate analysis results, and obtain yield estimates through an intuitive interface. The proposed solution is implemented as a research-oriented proof of

concept, demonstrating how AI-driven, image-based approaches can support precision agriculture in resource-constrained settings.

By focusing on accessibility, low-cost deployment, and ground-level imagery, this project addresses key challenges faced by Pakistani agriculture and offers a scalable framework that can be extended to other orchard crops. The proposed system highlights the potential of artificial intelligence to enhance data-driven decision-making, improve yield forecasting, and support sustainable agricultural practices in Pakistan.

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SYED AQIB SHAH & HAMZA AHMED
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"The goal of computer vision is to make machines see and understand the world as we do, so we can turn data into insight and insight into action."

Fei-Fei Li

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Acronyms and Abbreviations

AI	Artificial Intelligence
CV	Computer Vision
CNN	Convolutional Neural Network
DL	Deep Learning
FYP	Final Year Project
GUI	Graphical User Interface
HEC	Higher Education Commission (Pakistan)
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
ML	Machine Learning
RGB	Red, Green, Blue (Color Model)
UI	User Interface
YOLO	You Only Look Once (Object Detection Algorithm)
IoU	Intersection over Union
mAP	Mean Average Precision
API	Application Programming Interface
U-Net	Convolutional Network for Image Segmentation
GPS	Global Positioning System
CSV	Comma-Separated Values
PDF	Portable Document Format
APK	Android Package Kit
VVI	Vegetation Vigor Index
LCA	Leaf Coverage Area

Chapter 1

Introduction

1.1 Project Background & Motivation

Agriculture remains a fundamental pillar of the global economy, playing a critical role in food security, rural livelihoods, and economic stability. With the increasing global population and growing pressure on natural resources, there is a rising demand for more efficient, data-driven agricultural practices. Precision agriculture has emerged as a promising approach to address these challenges by leveraging digital technologies to optimize crop management, reduce waste, and improve productivity [1].

Fruit yield estimation is a key component of precision agriculture, particularly for orchard-based crops such as apples. Accurate pre-harvest yield estimates enable farmers to plan labor requirements, optimize harvesting schedules, manage storage and logistics, and make informed market decisions. However, traditional yield estimation methods typically rely on manual sampling, visual inspection, or farmer intuition, which are labor-intensive, time-consuming, and often unreliable, especially in large or heterogeneous orchards [2].

Recent advances in computer vision and deep learning have created new opportunities to automate yield estimation using image-based analysis. With the widespread availability of smartphones and low-cost cameras, it has become feasible to capture ground-level images of orchards and extract meaningful information such as fruit count, fruit maturity, and canopy structure. This project is motivated by the need to develop an accessible, low-cost, and scalable AI-based solution for fruit yield estimation that relies solely on ground-level imagery and does not require expensive sensors, drones, or historical yield records.

1.2 Significance of Yield Estimation in Agriculture

Accurate yield estimation plays a vital role across multiple stages of the agricultural value chain. For farmers, reliable yield forecasts support better decision-making related to irrigation, fertilization, pest management, harvesting, and labor allocation. At a broader scale, yield estimates contribute to supply chain planning, pricing strategies, food security assessments, and policy formulation [3].

In orchard systems, yield is influenced by multiple interacting factors, including fruit load, fruit development stage, and tree vigor. Estimating yield based solely on a small number of sample trees often fails to capture the spatial variability present within orchards, leading to inaccurate predictions [4]. This problem is particularly relevant for apple orchards, where occlusion, uneven fruit distribution, and canopy density complicate manual assessment.

Automated, image-based yield estimation offers a scalable alternative that can reduce subjectivity and improve consistency. By analyzing visual cues such as fruit count, color-based ripeness indicators, and canopy structure, AI-driven systems can provide more comprehensive and objective yield estimates. Such approaches are especially valuable in regions where access to advanced agricultural infrastructure is limited, making low-cost and image-based solutions highly significant.



(a) Dataset for Leave Area (b) Annotation according to Leave Area Coverage

Figure 1.1: Data Annotation for Leave Area Coverage Segmentation

1.3 Project Overview

This project proposes an AI-based fruit yield estimation system designed for apple orchards using only ground-level RGB images. The system integrates multiple computer vision components into a unified pipeline that estimates orchard-level yield in a practical and accessible manner.

The proposed framework consists of four core modules. First, a fruit detection and counting module identifies and localizes apples in ground-level images using deep learning-based object detection models. Second, a fruit ripeness classification module analyzes detected fruits and assigns them to discrete ripeness stages based on visual characteristics such as color and texture. Third,

a canopy feature extraction module segments tree canopies from the background and derives structural indicators such as projected canopy area and spread. Finally, a literature-informed yield estimation model combines fruit counts, ripeness distribution, and canopy features to generate approximate yield estimates without relying on locally measured yield ground truth.

To enhance usability, the system is integrated into a farmer-friendly mobile dashboard that allows users to capture images, view intermediate results, and obtain yield estimates through an intuitive interface. The project is implemented as a research-oriented proof of concept, demonstrating the feasibility of multimodal, image-based yield estimation using accessible technologies.

1.4 Role of Artificial Intelligence in Agriculture

Artificial intelligence has become a transformative force in modern agriculture, enabling automation, predictive analytics, and decision support across a wide range of applications. Machine learning and deep learning techniques have been successfully applied to crop monitoring, disease detection, weed identification, irrigation management, and yield prediction [5].

In the context of fruit yield estimation, deep learning-based computer vision models play a central role. Object detection architectures such as YOLO and Faster R-CNN have demonstrated strong performance in detecting fruits under challenging orchard conditions, including occlusions, variable lighting, and cluttered backgrounds [6, 4]. Similarly, convolutional neural networks have been used to classify fruit ripeness and quality based on visual features, providing valuable information for harvest timing and yield interpretation [7].

AI also enables the integration of multiple visual cues into unified predictive models. By combining fruit count, maturity indicators, and canopy characteristics, AI-based systems can capture complex relationships that are difficult to model using traditional statistical approaches. This project leverages these capabilities to design a lightweight, image-driven yield estimation framework suitable for low-resource and real-world agricultural settings.

1.5 Scalability and Future Scope

Although this project focuses on apple orchards as a case study, the proposed system architecture is designed to be modular and extensible. With appropriate training data, the fruit detection and ripeness classification modules can be adapted to other orchard crops such as citrus, mango, or peach. Similarly, the canopy segmentation and feature extraction approach can be generalized to different tree structures and planting patterns.

Future extensions of the system may include the integration of environmental data, such as temperature and rainfall, or temporal analysis using images captured at multiple growth stages to improve yield accuracy. Additionally, the system could be enhanced with calibration using locally collected yield data when available, transforming the proof-of-concept model into a fully supervised predictive framework.

The mobile dashboard may also be extended to support multilingual interfaces, offline inference, and cloud-based analytics, further increasing accessibility and adoption among farmers and agricultural stakeholders.

1.6 Problem Statement

Accurate yield estimation in apple orchards remains a challenging task due to the reliance on manual sampling, subjective judgment, and limited scalability of traditional methods. Existing practices are often unable to capture orchard-level variability and are prone to significant estimation errors, particularly in large or heterogeneous orchards [2].

While recent research has demonstrated the effectiveness of computer vision for fruit detection and canopy analysis, many existing approaches treat these components independently or rely on costly UAV and remote sensing data [8]. Furthermore, most data-driven yield prediction models require extensive historical yield records and environmental sensor data, which may not be available in many agricultural contexts.

There is a clear need for an integrated, low-cost, and image-based yield estimation approach that combines fruit count, ripeness information, and canopy features using only ground-level imagery. Addressing this gap forms the core problem addressed by this project.

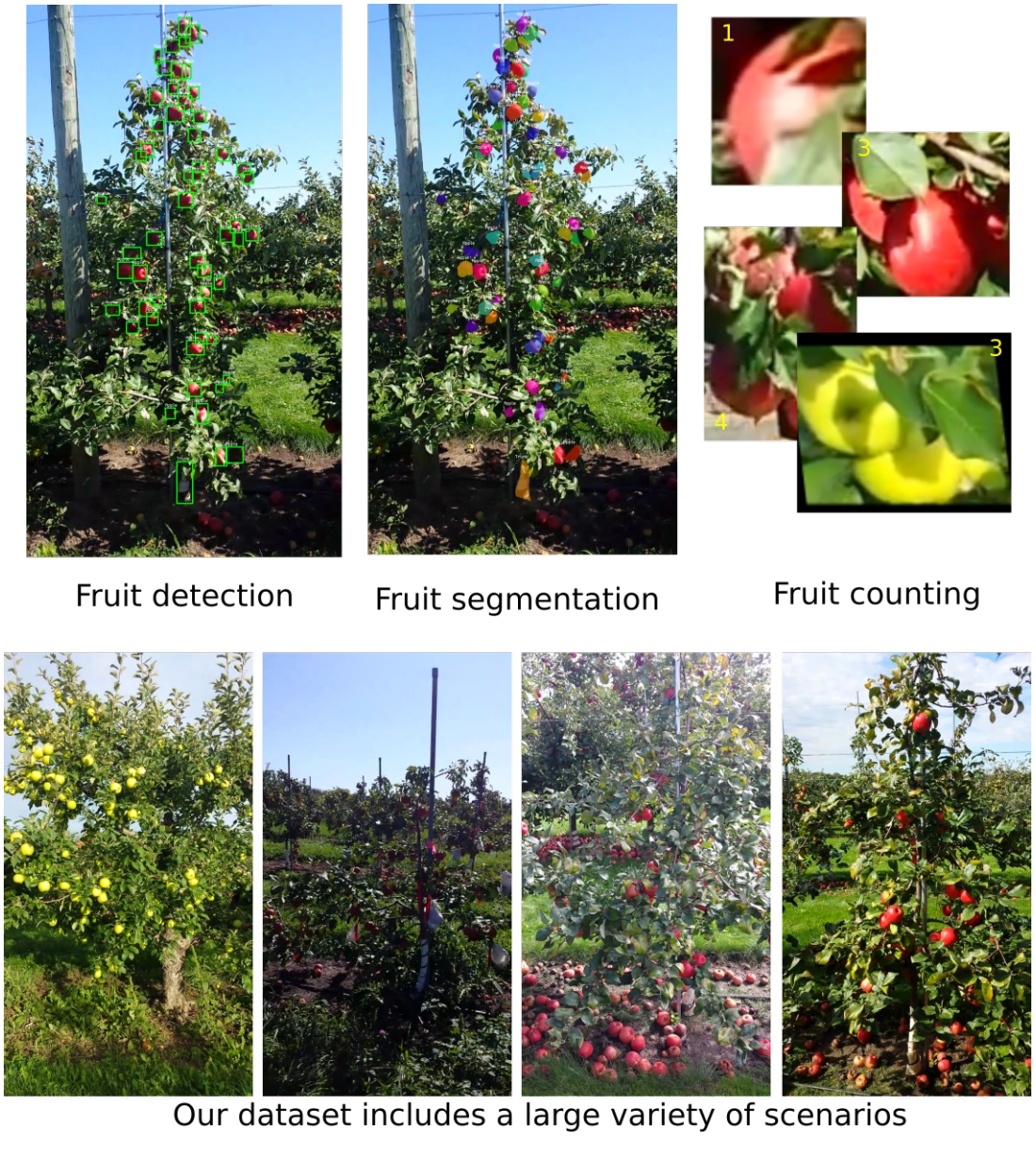


Figure 1.2: MinneApple Dataset for Fruit Counting

1.7 Objectives

The primary objectives of this project are as follows:

1. **Develop a Fruit Detection and Counting System**
   To design and train deep learning-based object detection models capable of accurately detecting and counting apples from ground-level orchard images.

2. **Implement Fruit Ripeness Classification**
   To develop a classification module that assigns detected fruits to discrete ripeness stages using image-based color and texture features.

3. **Extract Canopy Features from Ground-Level Images**
   To segment tree canopies and derive structural indicators such as projected canopy area and spread that reflect tree vigor.

4. **Design a Literature-Informed Yield Estimation Model**
   To combine fruit counts, ripeness distribution, and canopy features into a predictive model that estimates yield without relying on local ground-truth yield data.

5. **Develop a Farmer-Friendly Mobile Dashboard**
   To integrate the system into a mobile application that allows users to capture images and visualize fruit counts, ripeness summaries, canopy indicators, and estimated yield.

1.8 Scope of the Project

This project focuses on the development of an AI-based fruit yield estimation system using ground-level RGB images and publicly available datasets. The system is implemented as a research prototype and evaluated through controlled experiments.

In-Scope Features

- Fruit detection and counting using deep learning-based object detection models
- Image-based fruit ripeness classification
- Canopy segmentation and feature extraction from ground-level images
- Literature-informed yield estimation without local yield labels
- Mobile dashboard for visualization and user interaction

Out-of-Scope Features

- Use of UAV, satellite, or other remote-sensing imagery
- Integration of environmental or sensor-based data
- Physical deployment and large-scale field validation
- Economic forecasting or supply chain optimization

Chapter 2

Literature Review

2.1 Traditional Approaches to Fruit Yield Estimation

Traditional fruit yield estimation methods have primarily relied on manual sampling and visual inspection techniques, where a limited number of trees are selected and their fruit load is extrapolated to estimate orchard-level yield [2]. Although simple to implement, these approaches are labor-intensive, time-consuming, and highly subjective. They fail to account for spatial variability in orchards, differences in tree vigor, and variations in fruit distribution, leading to inaccurate yield estimates, particularly in large-scale or heterogeneous orchards [4].

2.2 Evolution of Precision Agriculture and Image-Based Yield Estimation

Precision agriculture has emerged as a data-driven approach aimed at optimizing agricultural productivity through targeted interventions and informed decision-making [1]. With advancements in sensing technologies and computing power, image-based analysis has become a key component of precision agriculture systems. Visual data captured from ground-level cameras or mobile devices enables scalable monitoring of crops and supports automated yield estimation without extensive manual effort [5].

2.3 Deep Learning for Fruit Detection in Orchards

Deep learning techniques, particularly convolutional neural networks (CNNs), have significantly advanced fruit detection in orchard environments. Object detection architectures such as Faster R-CNN and YOLO have demonstrated strong performance in localizing and counting fruits under complex outdoor conditions [9]. These models are capable of learning robust visual representations that handle cluttered backgrounds, varying illumination, and complex orchard structures, making them well-suited for automated fruit counting tasks [6].

2.4 Handling Occlusion, Overlapping Fruits, and Background Noise

Occlusion caused by leaves, branches, and overlapping fruits is one of the primary challenges in orchard fruit detection. Studies have shown that occlusion and background clutter significantly degrade detection accuracy if not addressed during model design and training [10]. Techniques such as multi-scale feature extraction, dense annotations, and extensive data augmentation have been proposed to improve robustness under occluded conditions [11]. These strategies enable models to better distinguish fruits from background elements and partially visible objects.

2.5 Fruit Ripeness and Maturity Estimation Using RGB Images

Beyond fruit counting, ripeness and maturity estimation has gained attention due to its importance in harvest planning and yield interpretation. Image-based approaches using RGB color and texture features have been successfully applied to classify fruits into discrete ripeness stages [7]. Recent deep learning models demonstrate that ripeness information can be extracted directly from detected fruit regions, providing additional insight into crop readiness and enhancing yield estimation frameworks [12].

2.6 Canopy Segmentation and Leaf Area Coverage Estimation

Canopy structure is closely linked to tree vigor, photosynthetic capacity, and yield potential. Canopy segmentation enables the extraction of features such as projected leaf area, canopy spread, and density, which serve as indicators of tree health and productive capacity [13]. Incorporating canopy features into yield estimation models has been shown to improve prediction accuracy compared to fruit-count-only approaches [8].

2.7 Image Segmentation Techniques for Canopy Extraction

Image segmentation plays a critical role in isolating canopy regions from complex orchard backgrounds. Deep learning-based segmentation models such as U-Net and DeepLabv3+ have been widely adopted due to their ability to perform pixel-wise classification in challenging outdoor environments [14]. These models outperform traditional thresholding and clustering methods, particularly under varying lighting conditions and partial occlusions common in orchard imagery.

2.8 Multimodal Fusion for Yield Estimation

Recent research emphasizes the importance of multimodal fusion for accurate yield estimation. Integrating fruit count, ripeness distribution, and canopy features allows models to capture complementary information that reflects both crop load and tree capacity [3]. Multimodal approaches

have consistently demonstrated improved robustness and predictive performance compared to single-feature models, particularly in complex agricultural settings [15].

2.9 Research Gaps and Motivation for the Proposed System

Despite significant advancements, several gaps remain in existing research. Many yield estimation systems rely on UAVs, multispectral sensors, or extensive historical yield data, limiting their applicability in low-resource environments [8]. Additionally, most studies address fruit detection, ripeness estimation, or canopy analysis in isolation rather than integrating them into a unified pipeline. These limitations motivate the proposed system, which focuses on ground-level RGB imagery and integrates multiple visual cues into a practical, accessible yield estimation framework.

2.10 Summary

This chapter reviewed traditional and modern approaches to fruit yield estimation, highlighting the role of deep learning in fruit detection, occlusion handling, ripeness classification, and canopy segmentation. The literature supports the integration of these components into a multimodal framework, which forms the basis of the proposed AI-based fruit yield estimation system.

Chapter 3

Methodology

3.1 Proposed Methodology

Accurate fruit yield estimation in orchards remains a challenging task due to spatial variability among trees, fruit occlusion, overlapping canopies, and the impracticality of destructive yield measurements at scale. To address these challenges, this study proposes an AI-driven, regression-based methodology that integrates computer vision techniques with agronomic domain knowledge. Similar hybrid approaches that combine visual feature extraction with regression modeling have demonstrated strong performance in orchard yield estimation tasks [16, 17].

Unlike approaches that rely solely on fruit counting, the proposed methodology estimates yield at the tree level by jointly modeling fruit count, leaf area coverage, vegetative vigor, fruit growth stage, and fruit density. Yield estimation is formulated as a supervised regression problem, enabling the learning of non-linear relationships between visual indicators and yield outcomes, as recommended in recent precision agriculture research [18].

Due to the absence of real tree-level yield labels, a pseudo-ground-truth generation strategy based on horticultural principles is adopted. The use of pseudo labels is a well-established practice in agricultural systems where destructive yield measurement is economically or practically infeasible [19, 20].

3.1.1 System Overview

The proposed system follows a modular processing pipeline that transforms ground-level orchard images into quantitative yield predictions. Modular architectures have been widely adopted in orchard monitoring systems due to their scalability, interpretability, and robustness to environmental variability [**?**]. The methodology begins with image acquisition and manual annotation, followed by feature extraction using deep learning models. The extracted features are fused into a structured dataset used to train a regression model for yield estimation.

The system supports both tree-level yield prediction and orchard-level yield aggregation, enabling practical deployment in real-world agricultural environments where only a subset of trees can be sampled [16].

3.1.2 Workflow Pipeline

1. **Image Acquisition and Manual Annotation**
   Ground-level RGB images of apple trees are captured under natural orchard conditions using handheld devices. Manual annotation is performed for fruit bounding boxes and leaf area coverage masks. Manual labeling is preferred for foliage-related features, as automated segmentation methods often struggle under dense canopies, occlusion, and variable illumination [21].

2. **Fruit Detection and Counting**
   Fruit detection is performed using a deep learning-based object detection model trained to identify apples under occlusion, overlapping fruits, and cluttered backgrounds. CNN-based detectors such as YOLO and Faster R-CNN have been extensively used for fruit detection in orchards due to their robustness and real-time inference capability [20, 22]. The total number of detected fruits per tree is computed as a primary yield-related feature.

3. **Leaf Area Coverage Segmentation**
   Instead of full canopy segmentation, the methodology focuses on leaf area coverage segmentation. Leaf area coverage is a strong indicator of photosynthetic capacity and tree vigor, both of which are closely linked to fruit development and yield potential [23]. A semantic segmentation model is trained using manually annotated masks to estimate the leaf coverage area for each tree.

4. **Vegetative Vigor Index Estimation**
   Tree vigor is quantified using a *Vegetative Vigor Index (VVI)* derived from leaf area coverage and color-based features extracted from RGB images. Vegetation vigor has been shown to correlate strongly with fruit retention, nutrient availability, and productivity in orchard systems [24]. The VVI is normalized to the range $VVI_i \in [0, 1]$, where higher values indicate healthier and more productive trees.

5. **Fruit Growth Stage Classification**
   Detected fruits are classified into Early, Mid, or Mature growth stages using RGB-based visual features such as size, color, and texture. Growth stage classification is widely used as a proxy for fruit weight estimation in orchard monitoring systems [17].

6. **Pseudo Ground-Truth Yield Generation**
   Due to the lack of real yield labels, pseudo ground truth is generated using horticultural and literature knowledge. Average fruit weights are assigned according to growth stage, following standard pomological practices [25]. The raw yield per tree is computed as:

$$Y_i^{raw} = \frac{F_i \times W_i}{1000}$$

   where F_i is the detected fruit count and W_i is the average fruit weight in grams.

To incorporate tree condition, the yield is adjusted using the Vegetative Vigor Index:

$$Y_i = Y_i^{raw} \times VVI_i$$

Such pseudo-labeling strategies are commonly employed in precision agriculture when destructive sampling is not feasible [19].

7. **Feature Vector Construction**
   For each tree, the following feature vector is constructed:

$$X_i = [FruitCount, VegetativeVigorIndex, GrowthStage, LeafCoverageArea, FruitDensity]$$

   Fruit density is computed as the ratio of fruit count to leaf coverage area. Feature fusion approaches have been shown to significantly improve yield estimation accuracy compared to single-feature models [18].

8. **Regression Model Training**
   Yield estimation is modeled as a supervised regression task:

$$f(X_i) \rightarrow Y_i$$

   Random Forest Regression is selected as the primary model due to its ability to model non-linear feature interactions and its proven effectiveness in agricultural yield prediction tasks [26].

9. **Model Evaluation and Explainability**
   Model performance is evaluated using Mean Squared Error (MSE). Feature importance analysis is conducted to quantify the contribution of each input feature. Explainable machine learning techniques such as feature importance and SHAP values are increasingly recommended for agricultural decision-support systems [27].

10. **Orchard-Level Yield Estimation**
    Predicted tree-level yields are aggregated to estimate orchard-level yield:

$$OrchardYield = \left(\frac{\sum_{i=1}^{N_{sample}} \hat{Y}_i}{N_{sample}} \right) \times N_{orchard}$$

    This sampling-based aggregation strategy is widely used in orchard-scale yield estimation studies [16].

3.1.3 Summary

This chapter presented a regression-based AI methodology for apple yield estimation using computer vision and agronomic knowledge. By integrating fruit detection, leaf area coverage segmentation,

vegetative vigor assessment, and growth stage analysis, the proposed system enables reliable yield prediction without requiring destructive ground-truth measurements. The adoption of pseudo-ground-truth generation enables supervised learning while maintaining practical feasibility, aligning with established practices in precision agriculture research [18, 19].

Chapter 4

Experimental Setup

4.1 Overview

The experimental setup for this study focuses on evaluating the proposed AI-driven apple yield estimation system. The experiments aim to validate both the **data collection and annotation process** and the **performance of computer vision models** used for fruit detection, leaf area coverage segmentation, and feature extraction for regression-based yield estimation. This chapter details the **experimental environment, datasets, hardware, software tools, and preliminary model training setups** employed in this project.

4.2 Experimental Environment

4.2.1 Hardware Configuration

Experiments were conducted on a high-performance workstation to ensure efficient model training and evaluation. The hardware specifications are as follows:

- **CPU:** Intel Core i9-13900K @ 3.0 GHz
- **GPU:** NVIDIA RTX A4000, 16 GB VRAM
- **RAM:** 64 GB DDR4
- **Storage:** 1 TB NVMe SSD

4.2.2 Software and Tools

The following software stack and libraries were utilized for development and experimentation:

- **Operating System:** Windows 10
- **Programming Language:** Python 3.11
- **Machine Learning Frameworks:** PyTorch, scikit-learn

- **Computer Vision Libraries:** OpenCV, Albumentations
- **Segmentation Models:** U-Net, U^2Net (for preliminary leaf area coverage segmentation)
- **Regression Models:** Random Forest Regressor, XGBoost

4.3 Dataset Preparation

4.3.1 Data Collection

The dataset consists of ground-level RGB images of apple trees captured under natural orchard conditions. Images were collected to represent a variety of:

- Tree sizes and canopy densities
- Fruit counts and spatial distribution
- Different growth stages (Early, Mid, Mature)

4.3.2 Manual Annotation

To create reliable ground truth for training segmentation models, manual annotation was performed for leaf area coverage only:

- **Leaf Area Coverage:** Pixel-level masks delineating visible leaf regions.

Fruit annotations were not manually created, as the MinneApple dataset already provides bounding box annotations for individual apples. Manual annotation of leaf regions ensured high-quality supervision for segmentation models, addressing challenges such as dense foliage, occlusion, and variable lighting conditions in real orchards.

4.3.3 Dataset Statistics

The annotated dataset currently comprises:

- **Images:** 950 annotated RGB images of apple trees
- **Annotated Fruits:** Approximately 9,500 individual fruit bounding boxes
- **Leaf Area Masks:** Pixel-wise annotations for specific trees

4.4 Model Training Setup

4.4.1 Leaf Area Coverage Segmentation

For preliminary segmentation, two deep learning models were trained on the manually annotated leaf area coverage masks:

- **U-Net:** Achieved an Intersection over Union (IoU) score of 0.68
- **U^{2}Net:** Achieved an IoU score of 0.71

Although these models provided acceptable baseline results, the segmentation quality leaves room for improvement. To enhance leaf area coverage accuracy, the next phase of experiments will employ **DeepLabv3+**, which has demonstrated superior performance in dense foliage segmentation tasks.

4.4.2 Feature Extraction

Following segmentation, relevant features are extracted per tree for the regression-based yield estimation pipeline:

- Fruit count per tree
- Leaf coverage area per tree
- Vegetative Vigor Index (VVI) computed from leaf area and color features
- Dominant fruit growth stage per tree
- Fruit density

These features form the input vectors for supervised regression models.

4.4.3 Regression Model Setup

Preliminary regression models are configured to predict tree-level yield in kilograms:

- **Random Forest Regressor:** Primary model due to its ability to capture non-linear interactions
- **XGBoost Regressor:** Alternative model for comparison
- **Training/Validation Split:** 80% training, 20% validation
- **Loss Metric:** Mean Squared Error (MSE)

The predicted tree-level yields will later be aggregated to generate orchard-level yield estimates.

Chapter 5

Another Chapter

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Chapter 6

Evaluation

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Chapter 7

Conclusions

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Appendix A

Appendix A

Appendices are provided to give supplementary information, which is included in the main text may serve as a distraction and cloud the central theme.

- Appendices should be numbered using alphabets, e.g. Appendix A, Appendix B, etc.
- Tables and References appearing in appendices should be numbered and referred to at appropriate places just as in the case of chapters.
- Appendices shall carry the title of the work reported and the same title shall be written in the contents page.

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